DWG. No. 310SC90-773

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REVISIONS

REV.	DESCRIPTION	REVD.	CHKD.	APPD.	DATE
B	To reflect the design progress & customer comments.	R.Kojima	M.Sakakibara	N.Matsumoto	2014-02-09
C	To reflect the design progress & customer comments.	R.Kojima	M.Sakakibara	N.Matsumoto	2014-02-18
D	To reflect the design progress & customer comments.	R.Kojima	M.Sakakibara	N.Matsumoto	2014-02-25
E	To reflect the design progress & customer comments.	R.Kojima	N.Matsumoto	N.Matsumoto	2014-06-18
F	To reflect the design progress & customer comments.	R.Kojima	N.Matsumoto	N.Matsumoto	2014-10-02
0	Submission as "Final" Engineering Document	R.Kojima	M.Sakakibara	N.Matsumoto	2015-01-13
1	To reflect Customer requirements	R.Kojima	Y.Kono	Y.Kono	2016-01-19
2	*To reflect Customer requirements*	*R.Kojima*	*Y.Kono*	*Y.Kono*	*2016-04-15*

INPUT/OUTPUT LIST BETWEEN CUSTOMER EQUIPMENT AND MHPS EQUIPMENT

三菱日立パワーシステムズ株式会社
2016-04-15
MITSUBISHI HITACHI POWER SYSTEMS, LTD.

FOR FINAL

CONFIDENTIAL

PROJECT TITLE / LOCATION	Drawn: Name/Date R. KOJIMA / 2013-05-15	Title, supplementary title	
SMC DAVAO POWER PLANT PROJECT MALITA, PROVINCE OF DAVAO DEL SUR, PHILIPPINES	Checked: Name/Date M. SAKAKIBARA / 2013-05-15	INPUT/OUTPUT LIST BETWEEN CUSTOMER EQUIPMENT AND MHPS EQUIPMENT	
	Approved: Name/Date N. MATSUMOTO / 2013-05-15		
FORMOSA HEAVY INDUSTRIES CORP. 100, SWEIKUAN RD, JENWU, KAOHSHIUNG, TAIWAN, R.O.C.	OWNER / ADDRESS	Responsible dep.	DWG. No. 310SC90-773 / Rev. 2 / Sheet 1/19 / Date of issue 2016-04-15
	San Miguel Consolidated Power Corporation 40 SAN MIGUEL AVENUE, MANDALUYONG CITY 1550 METRO MANILA, PHILIPPINES	Hitachi I&C Engineering Section/ Instrument & Control Department	MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division

CAD SOFTWARE: Excel2007 | SIZE DESIGNATION: A3

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1. Purpose

To operate the turbine safely and smoothly, we intent to clarify the hardware interface between FHI equipment and MHPS equipment in this drawing.

Hardware Interface is defined as the Input / Output (I/O) signals.

Input signals are defined as the signals from MHPS to FHI.

Output signals are defined as the signals from FHI to MHPS.

Hardware Interface is only applied for Control/Interlock/Protection/Alarm.

2. Scope of I/O List

We define the scope of Input/Output signals as following Figure.1.

For the detail of system configuration, please refer to Ref.1.

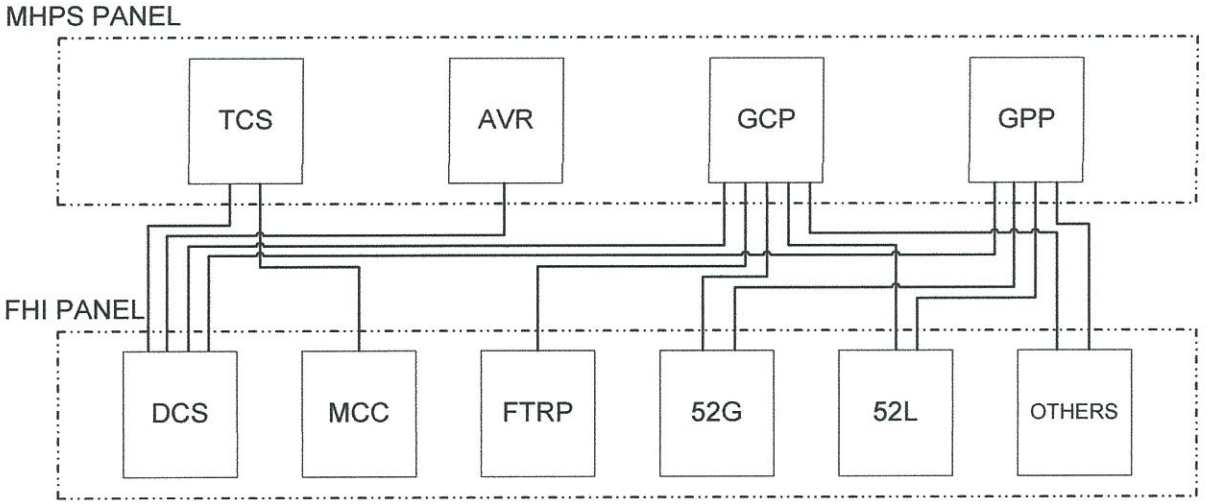


Figure. 1 Scope of I/O List

*1. Please provide MHPS the accurate panel name.
(Ref. 310SD03-743)

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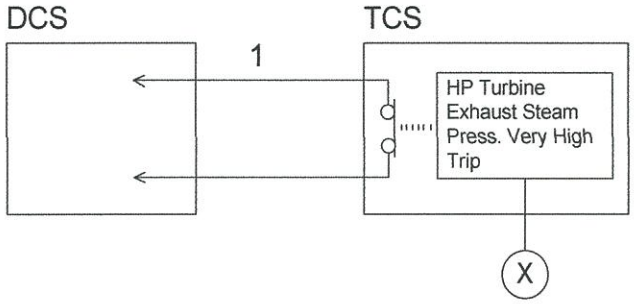
3. Remarks

3.1 Digital Signal Interface (DI/DO)

For all DI/DO signals, CONTACT MAKE means that the state of Control/Interlock/Protection is satisfied.

For example,

If the condition of "HP Turbine Exhaust Steam Press. Very High Trip" is satisfied, the contact of that signal is closed and the One state is output from TCS to DCS.



CONTACT MAKE = ONE(1) STATE

3.2 SOE

SOE is a part of DCS and is implemented in DCS.

4. Abbreviations:

TCS : TURBINE CONTROL SYSTEM
DCS : DISTRIBUTED CONTROL SYSTEM
TSI : TURBINE SUPERVISORY INSTRUMENT
GCP : GENERATOR CONTROL PANEL
GPP : GENERATOR PROTECTION RELAY PANEL
SOE : SEQUENCE OF EVENT RECORDER
FTRP : FULL TIME RECORDER PANEL
AVR : AUTOMATIC VOLTAGE REGULATOR CUBICLE
CCR : CENTRAL CONTROL ROOM
MHPS : MITSUBISHI HITACHI POWER SYSTEM

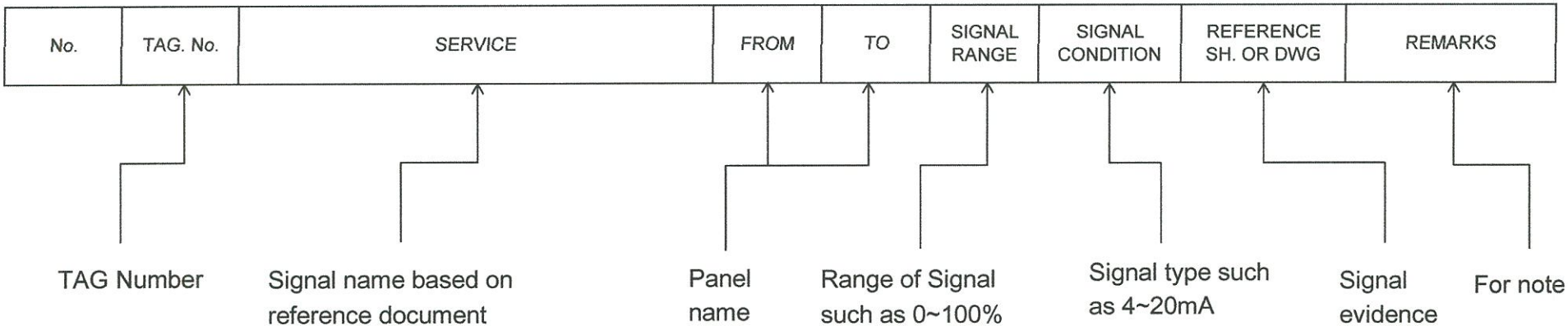
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5. Reference Documents and Drawings

Ref.1: Main Control System Configuration for Turbine/Generator and Auxiliaries (PCE-13-0104)
Ref.2: Turbine Generator Interlock Block Diagram (310SC90-774)
Ref.3: Generator Protection/Breaker Logic Diagram (Steam Turbine Generator) (310SD03-743)
Ref.4: One Line Diagram Generator Main Circuit (310SD03-741)
Ref.5: ATS Unit Sequence Control Program (310SC90-775)
Ref.6: ATS Criteria List and Criteria Table (310SC90-776)
Ref.7: Basic Configuration of Turbine Control System (TC-13-100)
Ref.8: Instrument List for Turbine and Generator (310SC90-779)
Ref.9: Wiring Diagram of Gen. Protection Panel, Gen Control and Metering Panel (GSE-2013-0082)
Ref.10: Turbine Local Control Diagram (310SD14-314)

6. Form sheet Explanation



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No.	TAG. No.	SERVICE	FROM	TO	SIGNAL RANGE	SIGNAL CONDITION	REFERENCE SH. OR DWG	REMARKS
1	S-TBMT	TURBINE MANUAL TRIP	TCS	DCS	—	DRY CONTACT	TA-1	FOR SOE
2	S-TBRESET	TURBINE RESET	TCS	DCS	—	DRY CONTACT	TA-1	FOR SOE
3	TCS-DI1003	TURBINE TRIP	TCS	DCS	—	DRY CONTACT	TA-1	FOR SOE
4	S-TMT	TCS MAJOR TROUBLE	TCS	DCS	—	DRY CONTACT	TA-9	FOR SOE
5	S-CVLL	CONDENSER VACUUM VERY LOW TRIP	TCS	DCS	—	DRY CONTACT	TA-10	FOR SOE
6	S-MSTLL	MAIN STEAM TEMP. VERY LOW	TCS	DCS	—	DRY CONTACT	TA-11	FOR SOE
7	S-TBFT	THRUST BEARING FAILURE (FRONT) ABNORMAL TRIP	TCS	DCS	—	DRY CONTACT	TA-12	FOR SOE
8	S-TBRT	THRUST BEARING FAILURE (REAR) ABNORMAL TRIP	TCS	DCS	—	DRY CONTACT	TA-12	FOR SOE
9	S-COPLL	CONTROL OIL PRESS. VERY LOW TRIP	TCS	DCS	—	DRY CONTACT	TA-13	FOR SOE
10	S-BOPLL	BEARING OIL PRESS. VERY LOW TRIP	TCS	DCS	—	DRY CONTACT	TA-13	FOR SOE
11	S-TBEHSTHH	LP TURBINE EXHAUST HOOD STEAM TEMP. VERY HIGH TRIP	TCS	DCS	—	DRY CONTACT	TA-14	FOR SOE
12	S-TBTOPLL	TURBINE TRIP OIL PRESS.VERY LOW TRIP	TCS	DCS	—	DRY CONTACT	TA-14	FOR SOE
13	S-TBBUOV	TURBINE BACK-UP OVERSPEED TRIP	TCS	DCS	—	DRY CONTACT	TA-15	FOR SOE
14	S-TBOT	TURBINE OVERSPEED TRIP	TCS	DCS	—	DRY CONTACT	TA-15	FOR SOE
15	S-TB1SVHH	NO.1 TURBINE SHAFT VIBRATION VERY HIGH	TCS	DCS	—	DRY CONTACT	TA-17	FOR SOE
16	S-TB2SVHH	NO.2 TURBINE SHAFT VIBRATION VERY HIGH	TCS	DCS	—	DRY CONTACT	TA-17	FOR SOE
17	S-G3SVHH	NO.3 GEN. SHAFT VIBRATION VERY HIGH	TCS	DCS	—	DRY CONTACT	TA-17	FOR SOE
18	S-G4SVHH	NO.4 GEN. SHAFT VIBRATION VERY HIGH	TCS	DCS	—	DRY CONTACT	TA-17	FOR SOE
19	S-TBESTHH	HP TURBINE EXHAUST STEAM TEMP. VERY HIGH TRIP	TCS	DCS	—	DRY CONTACT	TA-20	FOR SOE
20	S-TBIESDPHH	HP TURBINE INLET/EXHAUST STEAM DIFF. PRESS. VERY HIGH TRIP	TCS	DCS	—	DRY CONTACT	TA-21	FOR SOE

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21	S-TBESPHH	HP TURBINE EXHAUST STEAM PRESS. VERY HIGH TRIP	TCS	DCS	—	DRY CONTACT	TA-22	FOR SOE
22	S-PLURO	POWER LOAD UNBALANCE RELAY OPERATED	TCS	DCS	—	DRY CONTACT	TA-33	FOR SOE
23	S-ARO	ACCELERATION RELAY OPERATED	TCS	DCS	—	DRY CONTACT	TA-33	FOR SOE
24	TCS-DI1024	RSV/ICV FULLY CLOSED 1	TCS	DCS	—	DRY CONTACT	TA-6	FOR BOILER REHEATER PROTECTION
25	TCS-DI1025	RSV/ICV FULLY CLOSED 2	TCS	DCS	—	DRY CONTACT	TA-6	FOR BOILER REHEATER PROTECTION
26	TCS-DI1026	MSV/HPCV FULLY CLOSED AND TCS MAJOR TROUBLE 1	TCS	DCS	—	DRY CONTACT	TA-6	FOR BOILER REHEATER PROTECTION
27	TCS-DI1027	MSV/HPCV FULLY CLOSED AND TCS MAJOR TROUBLE 2	TCS	DCS	—	DRY CONTACT	TA-6	FOR BOILER REHEATER PROTECTION
28	TCS-DI1028	CV & ICV FULL CLOSED BY PLU ACTIVATION	TCS	DCS	—	DRY CONTACT	TA-6	FOR HOUSE LOAD OPERATION
29	TCS-DI1029	TURBINE SPEED UNDER 2400 RPM	TCS	DCS	—	DRY CONTACT	TA-15	FOR VACUUM UP
30		MAIN OIL PUMP (A) START	TCS	MCC	—	DRY CONTACT	TB-1	
31		MAIN OIL PUMP (B) START	TCS	MCC	—	DRY CONTACT	TB-2	
32		TURNING GEAR MOTOR START	TCS	MCC	—	DRY CONTACT	TB-9	
33	TCS-DI1030	TURNING GEAR ENGAGED	TCS	DCS	—	DRY CONTACT	TB-10	
34		MAIN OIL TANK VAPOUR EXTRACTOR START	TCS	MCC	—	DRY CONTACT	TB-12	
35		OIL PURIFIER PUMP START	TCS	MCC	—	DRY CONTACT	TB-15	
36		CONTROL OIL PUMP (A) START	TCS	MCC	—	DRY CONTACT	TB-21	
37		CONTROL OIL PUMP (B) START	TCS	MCC	—	DRY CONTACT	TB-22	
38		MSV (A) ABOVE SEAT SECONDARY DRAIN VALVE OPEN	TCS	MCC	—	DRY CONTACT	TC-1	
39		MSV (A) ABOVE SEAT SECONDARY DRAIN VALVE CLOSE	TCS	MCC	—	DRY CONTACT	TC-1	
40	TCS-DI1031	ICV TO CV TRANSFER COMPLETE	TCS	DCS	—	DRY CONTACT	TC-1	

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41		MSV (B) ABOVE SEAT SECONDARY DRAIN VALVE OPEN	TCS	MCC	—	DRY CONTACT	TC-2	
42		MSV (B) ABOVE SEAT SECONDARY DRAIN VALVE CLOSE	TCS	MCC	—	DRY CONTACT	TC-2	
43		CV (B) BELOW SEAT SECONDARY WARMING DRAIN VALVE OPEN	TCS	MCC	—	DRY CONTACT	TC-5	
44		CV (B) BELOW SEAT SECONDARY WARMING DRAIN VALVE CLOSE	TCS	MCC	—	DRY CONTACT	TC-5	
45		CV (A) ABOVE SEAT SECONDARY DRAIN VALVE OPEN	TCS	MCC	—	DRY CONTACT	TC-7	
46		CV (A) ABOVE SEAT SECONDARY DRAIN VALVE CLOSE	TCS	MCC	—	DRY CONTACT	TC-7	
47		CV (B) ABOVE SEAT SECONDARY DRAIN VALVE OPEN	TCS	MCC	—	DRY CONTACT	TC-8	
48		CV (B) ABOVE SEAT SECONDARY DRAIN VALVE CLOSE	TCS	MCC	—	DRY CONTACT	TC-8	
49		CRV (A) BELOW SEAT SECONDARY DRAIN VALVE OPEN	TCS	MCC	—	DRY CONTACT	TC-12	
50		CRV (A) BELOW SEAT SECONDARY DRAIN VALVE CLOSE	TCS	MCC	—	DRY CONTACT	TC-12	
51		CRV (B) BELOW SEAT SECONDARY DRAIN VALVE OPEN	TCS	MCC	—	DRY CONTACT	TC-13	
52		CRV (B) BELOW SEAT SECONDARY DRAIN VALVE CLOSE	TCS	MCC	—	DRY CONTACT	TC-13	
53		HP TURBINE WARMING VALVE OPEN	TCS	MCC	—	DRY CONTACT	TC-20	
54		HP TURBINE WARMING VALVE CLOSE	TCS	MCC	—	DRY CONTACT	TC-20	
55		GLAND STEAM EXHAUSTER START	TCS	MCC	—	DRY CONTACT	TC-28	
56		STEAM SEAL FEED INLET VALVE OPEN	TCS	MCC	—	DRY CONTACT	TC-31	
57		STEAM SEAL FEED INLET VALVE CLOSE	TCS	MCC	—	DRY CONTACT	TC-31	
58	TCS-DI1032	STEAM SEAL FEED INLET VALVE CLOSE	TCS	DCS	—	DRY CONTACT	TC-31	
59	TCS-DI1033	HP TURBINE WARMING LINE DRAIN VALVE OPEN COMMAND	TCS	DCS	—	DRY CONTACT	TE-1	
60	TCS-DI1034	HP TURBINE WARMING LINE DRAIN VALVE CLOSE COMMAND	TCS	DCS	—	DRY CONTACT	TE-1	

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61	TCS-DI1035	COLD REHEAT DRAIN VALVE (UPSTREAM OF C/R NRV) CLOSE COMMAND	TCS	DCS	—	DRY CONTACT	TE-1	
62	TCS-DI1036	COLD REHEAT DRAIN VALVE (UPSTREAM OF C/R NRV) FULL OPEN COMMAND	TCS	DCS	—	DRY CONTACT	TE-2	
63	TCS-DI1037	*TURBINE GLAND STEAM LINE DRAIN VALVE FULL OPEN* COMMAND	TCS	DCS	—	DRY CONTACT	TE-6	
64	TCS-DI1038	TURBINE GLAND STEAM LINE DRAIN VALVE FULL CLOSE COMMAND	TCS	DCS	—	DRY CONTACT	TE-6	
65		SPACE HEATER FOR GENERATOR START	TCS	MCC	—	DRY CONTACT	GD-1	
66		SPACE HEATER FOR AC EXCITER START	TCS	MCC	—	DRY CONTACT	GD-2	
67	TCS-DI1039	TURBINE START-UP AMBIENT COLD MODE	TCS	DCS	—	DRY CONTACT	310SC90-776 SH.20	FOR ATS
68	TCS-DI1040	TURBINE START-UP COLD 1 MODE	TCS	DCS	—	DRY CONTACT	310SC90-776 SH.20	FOR ATS
69	TCS-DI1041	TURBINE START-UP COLD 2 MODE	TCS	DCS	—	DRY CONTACT	310SC90-776 SH.20	FOR ATS
70	TCS-DI1042	TURBINE START-UP WARM MODE	TCS	DCS	—	DRY CONTACT	310SC90-776 SH.20	FOR ATS
71	TCS-DI1043	TURBINE START-UP HOT MODE	TCS	DCS	—	DRY CONTACT	310SC90-776 SH.20	FOR ATS
72	TCS-DI1044	TURBINE LOAD HOLD	TCS	DCS	—	DRY CONTACT	310SC90-776 SH.20	FOR ATS
~~73~~	~~TCS-DI1045~~	~~COORDINATION CONTROL PERMISSIVE CONDITION~~	~~TCS~~	~~DCS~~	~~—~~	~~DRY CONTACT~~	~~310SC90-775 SH.28~~	
74	*TCS-DI1046*	*DCS CONTROL MODE PERMISSIVE*	TCS	DCS	—	DRY CONTACT	TC-13-100	
75	TCS-DI1047	DCS CONTROL MODE ON FEEDBACK	TCS	DCS	—	DRY CONTACT	TC-13-100	
76	TCS-DI1048	MODBUS LINE-A DUTY	TCS	DCS	—	DRY CONTACT	TC-13-127	FOR NETWORK CHECKUP
77	TCS-DI1049	MODBUS LINE-B DUTY	TCS	DCS	—	DRY CONTACT	TC-13-127	FOR NETWORK CHECKUP
78	S-AMT	AVR MAJOR TROUBLE	AVR	DCS	—	DRY CONTACT	SH-G-4	FOR SOE
79	S-GLR86GAO	GEN. LOCKOUT RELAY (86G-A) OPERATED	GPP	DCS	—	DRY CONTACT	SH-G-5A	FOR SOE

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80	S-GLR86GBO	GEN. LOCKOUT RELAY (86G-B) OPERATED	GPP	DCS	—	DRY CONTACT	SH-G-6A	FOR SOE
81	S-GPD87G	PHASE DIFFERENTIAL OF GENENERATOR (87G)	GPP	DCS	—	DRY CONTACT	SH-G-8A	FOR SOE
82	S-GEF59GN	GEN. EARTH FAULT (59GN)	GPP	DCS	—	DRY CONTACT	SH-G-8A	FOR SOE
83	S-LOE40	LOSS OF EXCITATION (40)	GPP	DCS	—	DRY CONTACT	SH-G-8A	FOR SOE
84	S-NPS46	NEGATIVE-PHASE SEQ. (46)	GPP	DCS	—	DRY CONTACT	SH-G-8A	FOR SOE
85	S-PS78	POLE SLIP (78)	GPP	DCS	—	DRY CONTACT	SH-G-8A	FOR SOE
86	S-UF81L	UNDER FREQUENCY (81L, 2nd)	GPP	DCS	—	DRY CONTACT	SH-G-8A	FOR SOE
87	S-OF81H	OVER FREQUENCY (81H, 2nd)	GPP	DCS	—	DRY CONTACT	SH-G-8A	FOR SOE
88	S-VH24	VOLTS/HERTZ (24, 2nd or 3rd)	GPP	DCS	—	DRY CONTACT	SH-G-8A	FOR SOE
89	S-RP32	REVERSE POWER (32, 2nd)	GPP	DCS	—	DRY CONTACT	SH-G-8A	FOR SOE
90	S-OV59	OVER VOLTAGE (59, 2nd)	GPP	DCS	—	DRY CONTACT	SH-G-8A	FOR SOE
91	S-VRPOC51V	VOLTAGE RESTRAINED PHASE OVER CURRENT (51V)	GPP	DCS	—	DRY CONTACT	SH-G-8A	FOR SOE
92	S-HSPOC50	HIGH SET PHASE OVER CURRENT (50)	GPP	DCS	—	DRY CONTACT	SH-G-8A	FOR SOE
93	*S-UV27*	*UNDER VOLTAGE (27)*	*GPP*	*DCS*	—	*DRY CONTACT*	*SH-G-8A*	*FOR SOE*
94	S-BGPRIT	BOTH GEN. PROTECTION RELAY IN TROUBLE	GPP	DCS	—	DRY CONTACT	SH-G-8B	FOR SOE
95		52G CLOSE ORDER	GCP	52G	—	DRY CONTACT	SH-G-1B	
96		52L CLOSE ORDER	GCP	52L	—	DRY CONTACT	SH-G-2B	
97		52G OPEN BREAKER CS OPEN ORDER	SYN	52G	—	DRY CONTACT	SH-G-3	
98		52G OPEN BREAKER CS OPEN ORDER	SYN	52G	—	DRY CONTACT	SH-G-3	

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99		52L OPEN BREAKER CS OPEN ORDER	SYN	52L	—	DRY CONTACT	SH-G-3	
100		52L OPEN BREAKER CS OPEN ORDER	SYN	52L	—	DRY CONTACT	SH-G-3	
101		41 OPENED FOR ES INTERLOCK (GMCB)	AVR	OTHERS	—	DRY CONTACT	SH-G-4	
102		41 OPENED FOR 52G CLOSE INTERLOCK (AT LOCAL SIDE)	AVR	OTHERS	—	DRY CONTACT	SH-G-4	
103		41 OPENED FOR ES INTERLOCK	AVR	OTHERS	—	DRY CONTACT	SH-G-4	
104		GEN. PROTECTION OPERATED (52G-TC1)	GPP	52G	—	DRY CONTACT	SH-G-5B	
105		GEN. PROTECTION OPERATED (52G-TC2)	GPP	52G	—	DRY CONTACT	SH-G-5B	
106		GEN. PROTECTION OPERATED (52L-TC1)	GPP	52L	—	DRY CONTACT	SH-G-6B	
107		GEN. PROTECTION OPERATED (52L-TC2)	GPP	52L	—	DRY CONTACT	SH-G-6B	
108	MIT-GEN	GENERATOR TRIP	GPP	DCS	—	DRY CONTACT	SH-G-8B	
109	RY-52G	52G READY TO SYNCHRONIZE	GCP	DCS	—	DRY CONTACT	SH-G-8B	
110	RY-52L	52L READY TO SYNCHRONIZE	GCP	DCS	—	DRY CONTACT	SH-G-8B	
		GEN. LOCKOUT RELAY (86G-A) OPERATED	GPP	FTRP	—	DRY CONTACT	SH-G-5A	
		GEN. LOCKOUT RELAY (86G-B) OPERATED	*GPP*	*FTRP*	—	*DRY CONTACT*	*SH-G-6A*	

SERVICE	DIGITAL INPUT (6/6)			
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No.	TAG. No.	SERVICE	FROM	TO	SIGNAL RANGE	SIGNAL CONDITION	REFERENCE SH. OR DWG	REMARKS
1		TURBINE MANUAL TRIP (CCR) 1	CCR	TCS	—	DRY CONTACT	TA-1	
2		TURBINE MANUAL TRIP (CCR) 2	CCR	TCS	—	DRY CONTACT	TA-1	
3	BY FHI	BOILER MFT 1	DCS	TCS	—	DRY CONTACT	TA-9	Trip = 1
4	BY FHI	BOILER MFT 2	DCS	TCS	—	DRY CONTACT	TA-9	Trip = 1
5		MAIN OIL PUMP (A) STARTED	MCC	TCS	—	DRY CONTACT	TB-1	
6		MAIN OIL PUMP (A) MOTOR ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	TB-1	
7		MAIN OIL PUMP (B) STARTED	MCC	TCS	—	DRY CONTACT	TB-2	
8		MAIN OIL PUMP (B) MOTOR ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	TB-2	
9		TURNING GEAR MOTOR STARTED	MCC	TCS	—	DRY CONTACT	TB-9	
10		TURNING GEAR MOTOR MOTOR ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	TB-9	
11		MAIN OIL TANK VAPOUR EXTRACTOR STARTED	MCC	TCS	—	DRY CONTACT	TB-12	
12		MAIN OIL TANK VAPOUR EXTRACTOR MOTOR ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	TB-12	
13		OIL PURIFIER PUMP STARTED	MCC	TCS	—	DRY CONTACT	TB-15	
14		*OIL PURIFIER PUMP MOTOR ELECTRICAL TROUBLE*	*MCC*	*TCS*	—	*DRY CONTACT*	*TB-15*	
15		CONTROL OIL PUMP (A) STARTED	MCC	TCS	—	DRY CONTACT	TB-21	
16		CONTROL OIL PUMP (A) MOTOR ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	TB-21	
17		CONTROL OIL PUMP (B) STARTED	MCC	TCS	—	DRY CONTACT	TB-22	
18		CONTROL OIL PUMP (B) MOTOR ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	TB-22	
19		MSV (A) ABOVE SEAT SECONDARY DRAIN VALVE MOTOR ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	TC-1	
20		MSV (B) ABOVE SEAT SECONDARY DRAIN VALVE MOTOR ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	TC-2	

No.	TAG. No.	SERVICE	FROM	TO	SIGNAL RANGE	SIGNAL CONDITION	REFERENCE SH. OR DWG	REMARKS
21		CV (B) BELOW SEAT SECONDARY WARMING DRAIN VALVE MOTOR ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	TC-5	
22		CV (A) ABOVE SEAT SECONDARY DRAIN VALVE MOTOR ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	TC-7	
23		CV (B) ABOVE SEAT SECONDARY DRAIN VALVE MOTOR ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	TC-8	
24		CRV (A) BELOW SEAT SECONDARY DRAIN VALVE MOTOR ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	TC-12	
25		CRV (B) BELOW SEAT SECONDARY DRAIN VALVE MOTOR ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	TC-13	
26		HP TURBINE WARMING VALVE MOTOR ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	TC-20	
27		GLAND STEAM EXHAUSTER STARTED	MCC	TCS	—	DRY CONTACT	TC-28	
28		GLAND STEAM EXHAUSTER MOTOR ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	TC-28	
29		STEAM SEAL FEED INLET VALVE MOTOR ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	TC-31	
30								
31	LT-HD1101-1,2	NO.1 LP HEATER LEVEL VERY HIGH	DCS	TCS	—	DRY CONTACT	TC-36	
32	LT-HD1105-1,2	NO.1 HP HEATER LEVEL VERY HIGH	DCS	TCS	—	DRY CONTACT	TC-41	
33	LT-CD1104-1,2,3	DEAERATOR STORAGE TANK LEVEL VERY HIGH	DCS	TCS	—	DRY CONTACT	TC-42	
34	LT-HD1103-1,2	NO.3 LP HEATER LEVEL VERY HIGH	DCS	TCS	—	DRY CONTACT	TC-44	
35	LT-HD1102-1,2	NO.2 LP HEATER LEVEL VERY HIGH	DCS	TCS	—	DRY CONTACT	TC-45	
36	AUTO-AS1394	HP TURBINE WARMING LINE DRAIN VALVE AUTO SELECTED	DCS	TCS	—	DRY CONTACT	TE-1	Auto = 1, Manual = 0
37	AUTO-CR1011	COLD REHEAT DRAIN VALVE (UPSTREAM OF C/R NRV) AUTO SELECTED	DCS	TCS	—	DRY CONTACT	TE-1	Auto = 1, Manual = 0
38	DOZSO-AS1394	HP TURBINE WARMING LINE DRAIN VALVE OPEN ANSWER BACK	DCS	TCS	—	DRY CONTACT	TE-1	
39	DOZSC-AS1394	HP TURBINE WARMING LINE DRAIN VALVE CLOSE ANSWER BACK	DCS	TCS	—	DRY CONTACT	TE-1	
40	DOZSC-CR1011	COLD REHEAT DRAIN VALVE (UPSTREAM OF C/R NRV) CLOSE ANSWER BACK	DCS	TCS	—	DRY CONTACT	TE-1	

SERVICE: DIGITAL OUTPUT (2/6)

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No.	TAG. No.	SERVICE	FROM	TO	SIGNAL RANGE	SIGNAL CONDITION	REFERENCE SH. OR DWG	REMARKS
41	DOZSO-CR1011	COLD REHEAT DRAIN VALVE (UPSTREAM OF C/R NRV) FULL OPEN ANSWER BACK	DCS	TCS	—	DRY CONTACT	TE-2	
42	DOZSO-AV1391	CONDENSER VACUUM BREAK COMPLETE	DCS	TCS	—	DRY CONTACT	TE-6	
43	DOZSO-AS1392	*TURBINE GLAND STEAM LINE DRAIN VALVE FULL OPEN* ANSWER BACK	DCS	TCS	—	DRY CONTACT	TE-6	
44	DOZSC-AS1392	TURBINE GLAND STEAM LINE DRAIN VALVE FULL CLOSE ANSWER BACK	DCS	TCS	—	DRY CONTACT	TE-6	
45	TCS-DO1001	TURBINE GLAND STEAM LINE DRAIN VALVE AUTO	DCS	TCS	—	DRY CONTACT	TE-8	
46		SPACE HEATER FOR GENERATOR STARTED	MCC	TCS	—	DRY CONTACT	GD-1	
47		SPACE HEATER FOR AC EXCITER STARTED	MCC	TCS	—	DRY CONTACT	GD-2	
48	DOZSC-MS1020	HP BYPASS VALVE FULL CLOSE	DCS	TCS	—	DRY CONTACT	310SC90-776 SH-6	FOR ATS
49	AUTO-MS1020/ AUTO-HR1010	HP / LP BYPASS CONTROL IN AUTO MODE	DCS	TCS	—	DRY CONTACT	310SC90-776 SH-6	FOR ATS
50	TCS-DO1002	DCS CONTROL MODE ON	DCS	TCS	—	DRY CONTACT	TC-13-100	Pulse signal
51	TCS-DO1003	DCS CONTROL MODE OFF	DCS	TCS	—	DRY CONTACT	TC-13-100	Pulse signal
52	TCS-DO1004	MODBUS LINE-A PERMISSION	DCS	TCS	—	DRY CONTACT	TC-13-127	FOR NETWORK CHECKUP
53	TCS-DO1005	MODBUS LINE-B PERMISSION	DCS	TCS	—	DRY CONTACT	TC-13-127	FOR NETWORK CHECKUP
54		52G OPENED	52G	TCS	—	DRY CONTACT	SH-G-1A	INCLUDES FOR ATS AND INDICATION
55		52G CLOSED	52G	TCS	—	DRY CONTACT	SH-G-1A	INCLUDES FOR ATS
56		52L OPENED (FOR HOUSE LOAD OPERATION)	52L	TCS	—	DRY CONTACT	SH-G-2A	
57		52L OPENED	52L	TCS	—	DRY CONTACT	SH-G-2A	INCLUDES FOR INDICATION
58		52G REMOTE SELECTED	52G	GCP	—	DRY CONTACT	SH-G-1A	
59		ES AND 52G OPENED	52G	AVR	—	DRY CONTACT	SH-G-1A	

SERVICE: DIGITAL OUTPUT (3/6)

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No.	TAG. No.	SERVICE	FROM	TO	SIGNAL RANGE	SIGNAL CONDITION	REFERENCE SH. OR DWG	REMARKS
60		DS CLOSED	52G	TCS	—	DRY CONTACT	SH-G-1A	
~~61~~		~~52G CLOSED~~	~~52G~~	~~TCS~~	—	~~DRY CONTACT~~	~~SH-G-1A~~	
62		52G CLOSED	52G	GPP	—	DRY CONTACT	SH-G-1A	
63		52G CLOSED	52G	GPP	—	DRY CONTACT	SH-G-1A	
64		52G CLOSED	52G	AVR	—	DRY CONTACT	SH-G-1A	
65		52G CLOSED	52G	SYN	—	DRY CONTACT	SH-G-1A	FOR INDICATION
66		52G OPENED	52G	SYN	—	DRY CONTACT	SH-G-1A	FOR INDICATION
67		52G CLOSED	52G	GCP	—	DRY CONTACT	SH-G-1A	
~~68~~		~~52G OPENED~~	~~52G~~	~~TCS~~	—	~~DRY CONTACT~~	~~SH-G-1A~~	
69		52L REMOTE SELECTED	52L	GCP	—	DRY CONTACT	SH-G-2A	
70		52L CLOSE PERMIT CONDITION	52L	GCP	—	DRY CONTACT	SH-G-2A	
71		52L CLOSED	52L	GCP	—	DRY CONTACT	SH-G-2A	
72		52L CLOSED	52L	SYN	—	DRY CONTACT	SH-G-2A	FOR INDICATION
73		52L OPENED	52L	SYN	—	DRY CONTACT	SH-G-2A	FOR INDICATION
74		52L OPENED	52L	GPP	—	DRY CONTACT	SH-G-2A	
~~75~~		~~52L OPENED (FOR INDICATION)~~	~~52L~~	~~TCS~~	—	~~DRY CONTACT~~	~~SH-G-2A~~	
~~76~~		~~52L OPENED (FOR HOUSE LOAD OPERATION)~~	~~52L~~	~~TCS~~	—	~~DRY CONTACT~~	~~SH-G-2A~~	
77		52L OPENED	52L	GPP	—	DRY CONTACT	SH-G-2A	
78		52L CLOSE PERMIT CONDITION FOR FIRST POWER RECEIVE	OTHERS	GCP	—	DRY CONTACT	SH-G-2B	
79	TCS-DO1006	AUTO SYN START PB ON	DCS	TCS	—	DRY CONTACT	SH-G-3	Pulse Signal

SERVICE: DIGITAL OUTPUT (4/6)

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No.	TAG. No.	SERVICE	FROM	TO	SIGNAL RANGE	SIGNAL CONDITION	REFERENCE SH. OR DWG	REMARKS
80		230kV SWYD LOCKOUT RELAY OPERATED	OTHERS	GCP	—	DRY CONTACT	SH-G-7	
81		50BF-86BF	OTHERS	GCP	—	DRY CONTACT	SH-G-7	
82		GEN. TRANSF. PROTECTION OPERATED	OTHERS	GCP	—	DRY CONTACT	SH-G-7	
83		UNIT. TRANSF. PROTECTION OPERATED	OTHERS	GCP	—	DRY CONTACT	SH-G-7	
84		50BF-86BF FOR GEN. PROTECTION A	OTHERS	GPP	—	DRY CONTACT	SH-G-7	
85		GEN. TRANSF. PROTECTION OPERATED FOR GEN. PROTECTION A	OTHERS	GPP	—	DRY CONTACT	SH-G-7	
86		UNIT. TRANSF. PROTECTION OPERATED FOR GEN. PROTECTION A	OTHERS	GPP	—	DRY CONTACT	SH-G-7	
87		50BF-86BF FOR GEN. PROTECTION A	OTHERS	TCS	—	DRY CONTACT	SH-G-7	
88		GEN. TRANSF. PROTECTION OPERATED FOR GEN. PROTECTION A	OTHERS	TCS	—	DRY CONTACT	SH-G-7	
89		UNIT. TRANSF. PROTECTION OPERATED FOR GEN. PROTECTION A	OTHERS	TCS	—	DRY CONTACT	SH-G-7	
90		50BF-86BF FOR GEN. PROTECTION B	OTHERS	GPP	—	DRY CONTACT	SH-G-7	
91		GEN. TRANSF. PROTECTION OPERATED FOR GEN. PROTECTION B	OTHERS	GPP	—	DRY CONTACT	SH-G-7	
92		UNIT. TRANSF. PROTECTION OPERATED FOR GEN. PROTECTION B	OTHERS	GPP	—	DRY CONTACT	SH-G-7	
93		*50BF-86BF FOR GEN. PROTECTION B*	*OTHERS*	*TCS*	—	*DRY CONTACT*	*SH-G-7*	
94		GEN. TRANSF. PROTECTION OPERATED FOR GEN. PROTECTION B	OTHERS	TCS	—	DRY CONTACT	SH-G-7	
95		UNIT. TRANSF. PROTECTION OPERATED FOR GEN. PROTECTION B	OTHERS	TCS	—	DRY CONTACT	SH-G-7	
96		GEN. TRANSF. PROTECTION OPERATED	OTHERS	TCS	—	DRY CONTACT	SH-G-7	FOR ANN
97		UNIT. TRANSF. PROTECTION OPERATED	OTHERS	TCS	—	DRY CONTACT	SH-G-7	FOR ANN
98	TCS-DI1050	HP-CV FULL CLOSE (BOTH CV (A) & (B))	TCS	DCS	—	DRY CONTACT	TA-6	
99	TCS-DI1051-1	MSV (A) TEST	TCS	DCS	—	DRY CONTACT	TA-31	

SERVICE: DIGITAL OUTPUT (5/6)

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No.	TAG. No.	SERVICE	FROM	TO	SIGNAL RANGE	SIGNAL CONDITION	REFERENCE SH. OR DWG	REMARKS
100	TCS-DI1051-2	MSV (B) TEST	TCS	DCS	—	DRY CONTACT	TA-32	
101		41 CLOSED	TCS	DCS	—	DRY CONTACT	G-4	The left SH.No. is only for understanding.
102		GEN. LOCKOUT RELAY (86G) "A" OPERATED	TCS	DCS	—	DRY CONTACT	G-5A	The left SH.No. is only for understanding.
103		GEN. LOCKOUT RELAY (86G) "B" OPERATED	TCS	DCS	—	DRY CONTACT	G-6A	The left SH.No. is only for understanding.
104		TURBINE TRIP	TCS	DCS	—	DRY CONTACT	TA-6	
		SPACE HEATER FOR GEN.ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	GD-1	
		SPACE HEATER FOR AC EXCITER. ELECTRICAL TROUBLE	MCC	TCS	—	DRY CONTACT	GD-2	

SERVICE	DIGITAL OUTPUT (6/6)			
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No.	TAG. No.	SERVICE	FROM	TO	SIGNAL RANGE	SIGNAL CONDITION	REFERENCE SH. OR DWG	REMARKS
1	TCS-AI1001	TARGET MW SETPOINT FEEDBACK	TCS	DCS	0～200 MW	4～20mA	TC-13-100	
2	TE-LOn002-1	LUBE OIL COOLER OUTLET OIL TEMP. (A)	TCS	DCS	0～100degC	4～20mA	TD-1	FOR TEMP. CONTROL
3	TE-LOn002-2	LUBE OIL COOLER OUTLET OIL TEMP. (B)	TCS	DCS	0～100degC	4～20mA	TD-1	FOR TEMP. CONTROL
4	TCS-AI1004	TURBINE RATED SPEED	TCS	DCS	0～4000rpm	4～20mA	TD-1	
5	TCS-AI1005	TURBINE LOAD UP RATE	TCS	DCS	0～5%/MIN	4～20mA	SH-20	FOR ATS
6	GCP-AI1001	GEN. MW (C)	GCP	DCS	0～200 MW	4～20mA	310SD03-741	FOR CUSTOMER USE
7	GCP-AI1002	GEN. MW(D)	GCP	DCS	0～200 MW	4～20mA	310SD03-741	FOR CUSTOMER USE
8		GEN. VOLTAGE	GCP	FTRP	0～13.8 KV	0～115V	310SD03-741	FOR CUSTOMER USE
9	GCP-AI1003	GEN. WATT-HOUR	GCP	DCS	1000kWh/Pulse	Pulse(dry contact)	GSE-2013-0082	FOR CUSTOMER USE

SERVICE	ANALOG INPUT			
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No.	TAG. No.	SERVICE	FROM	TO	SIGNAL RANGE	SIGNAL CONDITION	REFERENCE SH. OR DWG	REMARKS
1	TCS-AO1001	TARGET MW DEMAND	DCS	TCS	0～200 MW	4～20mA	TC-13-100	
2	TCS-AS1004	AUX. STEAM HEADER PRESS. (LOWER)	DCS	TCS	0～15kg/cm^2(gauge)	4～20mA	TC-20	
3	ZT-AS1011	HP TURBINE WARMING VALVE POSITION	TCS	DCS	0～100 %	4～20mA	310SC90-779	
4	PT-ST1001-1	HP TURBINE FIRST STAGE STEAM PRESS. (A)	TCS	DCS	0～150 kg/cm2g	4～20mA	310SC90-779	
5	PT-ST1001-2	HP TURBINE FIRST STAGE STEAM PRESS. (B)	TCS	DCS	0～150 kg/cm2g	4～20mA	310SC90-779	
6	PT-ST1001-3	HP TURBINE FIRST STAGE STEAM PRESS. (C)	TCS	DCS	0～150 kg/cm2g	4～20mA	310SC90-779	
7	PV-GS1031	STEAM SEAL FEED VALVE MANIPULATION VALUE	TCS	DCS	0～100 %	4～20mA	310SD14-314	The left SH.No. is only for understanding.
8		GEN. MVAR	TCS	DCS	Lead 240～0～Lag 240 MVar	4～20mA	GSE-2013-0082	The left SH.No. is only for understanding.
9		GEN. CURRENT (A)	TCS	DCS	0～10 KA	4～20mA	GSE-2013-0082	The left SH.No. is only for understanding.
10		GEN. VOLTAGE (A) - (B)	TCS	DCS	0～18 KV	4～20mA	GSE-2013-0082	The left SH.No. is only for understanding.
11		GEN. POWER FACTOR	TCS	DCS	Lead 0.5～1～Lag 0.5	4～20mA	GSE-2013-0082	The left SH.No. is only for understanding.
12		FREQUENCY	TCS	DCS	55～65	4～20mA	GSE-2013-0082	The left SH.No. is only for understanding.
		GEN. CURRENT (B)	TCS	DCS	0～10 KA	4～20mA	GSE-2013-0082	The left SH.No. is only for understanding.
		GEN. VOLTAGE (B) - (C)	TCS	DCS	0～18 KV	4～20mA	GSE-2013-0082	The left SH.No. is only for understanding.
		GEN. CURRENT (C)	TCS	DCS	0～10 KA	4～20mA	GSE-2013-0082	The left SH.No. is only for understanding.
		GEN. VOLTAGE (C) - (A)	TCS	DCS	0～18 KV	4～20mA	GSE-2013-0082	The left SH.No. is only for understanding.

SERVICE: ANALOG OUTPUT

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